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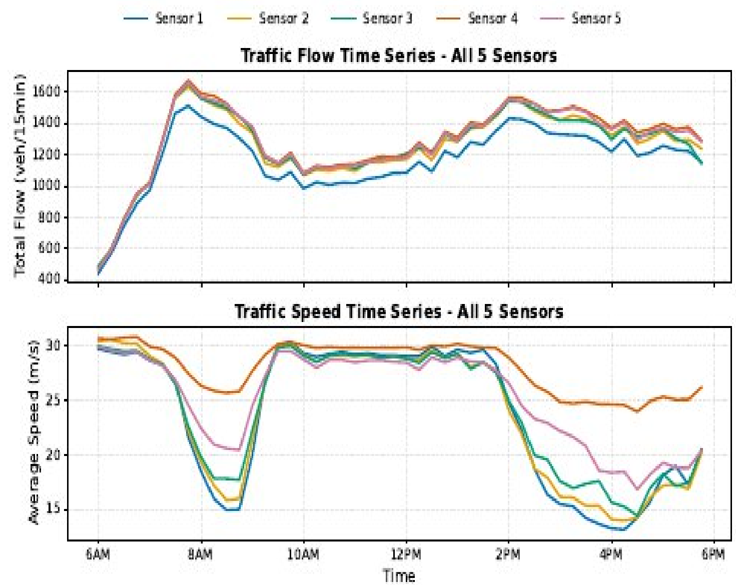
Problem & Motivation

Traffic simulation is critical for:

- Urban planning and infrastructure design
- Answering "what-if" questions
- Evaluating disruption scenarios



The Problem:

- Car-following models operate at **lane level**
- But validation uses **aggregated corridor metrics**
- Lane-specific dynamics are **hidden**

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Research Questions

RQ1: Does the choice of **numerical integrator** affect simulation accuracy?

RQ2: Does the **arrival process** (how vehicles enter) affect accuracy?

RQ3: What do **lane-level validation metrics** reveal that aggregate metrics hide?

Spoiler: Integrators don't matter much. Arrivals matter a lot.

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Car-Following: Intelligent Driver Model

The IDM computes acceleration as a continuous function:

$$a_n = a_{\max} \left[1 - \left(\frac{v_n}{v_{\max}} \right)^{\delta} - \left(\frac{s^*(v_n, \Delta v)}{s_n} \right)^2 \right]$$

Key properties:

- Smooth transitions: free-flow $\rightarrow$ following $\rightarrow$ braking
- No abrupt acceleration jumps (unlike Gipps model)
- 5 parameters: s_0, $a_{\max}$, b, T, $v_{\max}$

We used standard literature values (no calibration in this study)

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Arrival Process: Shifted Erlang-2

Problem with Poisson arrivals:

- Allows arbitrarily small headways (unrealistic)
- No minimum spacing between vehicles

Our solution: Shifted Erlang-2 distribution

$$Y = \tau - \mu \ln(U_1 U_2)$$

where $\tau > 0 =$ minimum feasible headway

Data-driven parameterization:

- $\bar{\mu}_{\ell,t} =$ mean inter-arrival from PeMS flow data
- $\mu = \frac{\bar{\mu}_{\ell,t} - \tau}{2}$ (matches empirical mean)

Enforces realistic spacing while matching observed flow

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Study Corridor: US-101 Northbound

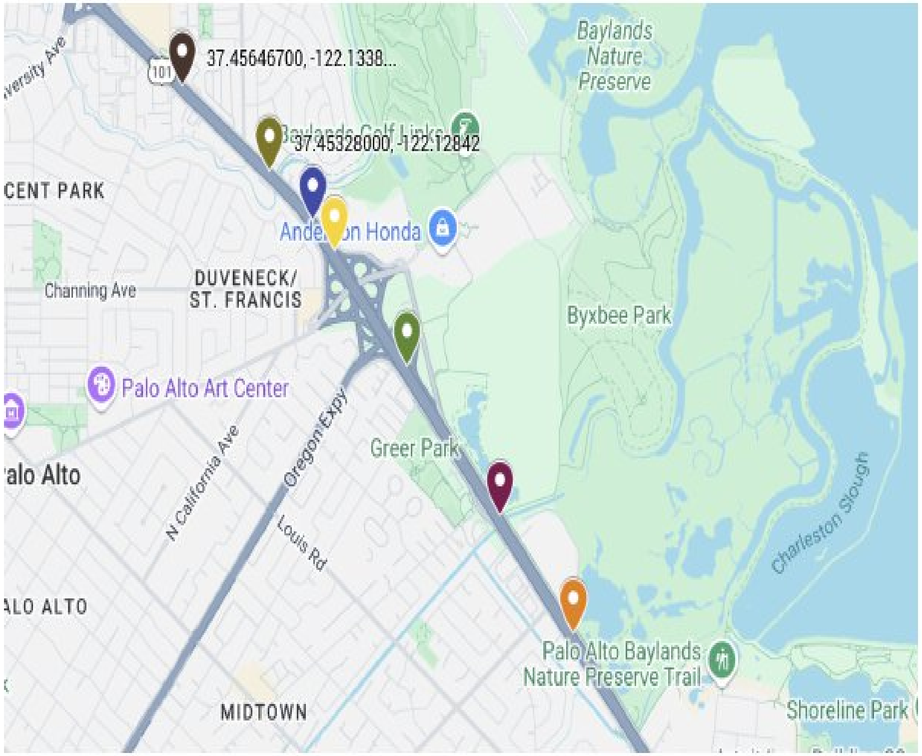
Location: Donald Doyle Highway, California

Characteristics:

- 1.4 km segment
- 4 mainline lanes
- 5 loop detector stations
- 2 on-ramp merge points

Data:

- PeMS loop detector data
- 12-hour period (6 AM–6 PM)
- 15-minute aggregation intervals
- ∼60,000 vehicles total



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Result 1: Numerical Integrators

Tested 8 integration methods:

- Euler, Heun, RK2, RK3, RK4
- Dormand-Prince (DOPRI5)
- Ballistic (simple kinematics)

Key Finding:

- Accuracy variation: $< \mathbf{1\%}$
- Runtime: Ballistic is **3**× **faster**

Recommendation: Use Ballistic

Method	Fitness	Time
Ballistic	0.157	17 min
RK4	0.157	51 min
Euler	0.158	44 min

Fitness = *combined flow* + *speed error*

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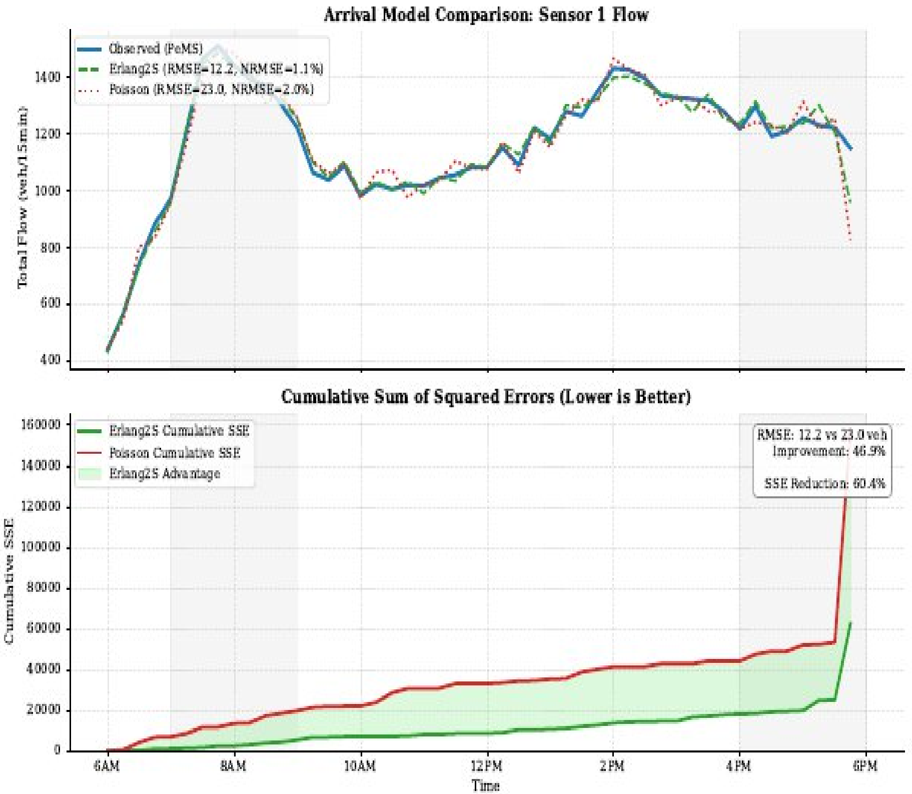
Result 2: Arrival Process Matters

Shifted Erlang-2 vs Poisson:

- Overall: **28% more accurate**
- Sensor 1 RMSE: **47% reduction**
  - Poisson: 23.04 vehicles
  - Erlang: 12.24 vehicles



Why?

- Minimum headway (τ) prevents unrealistic bunching
- Gives IDM proper spacing to work with

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Lane-specific patterns:

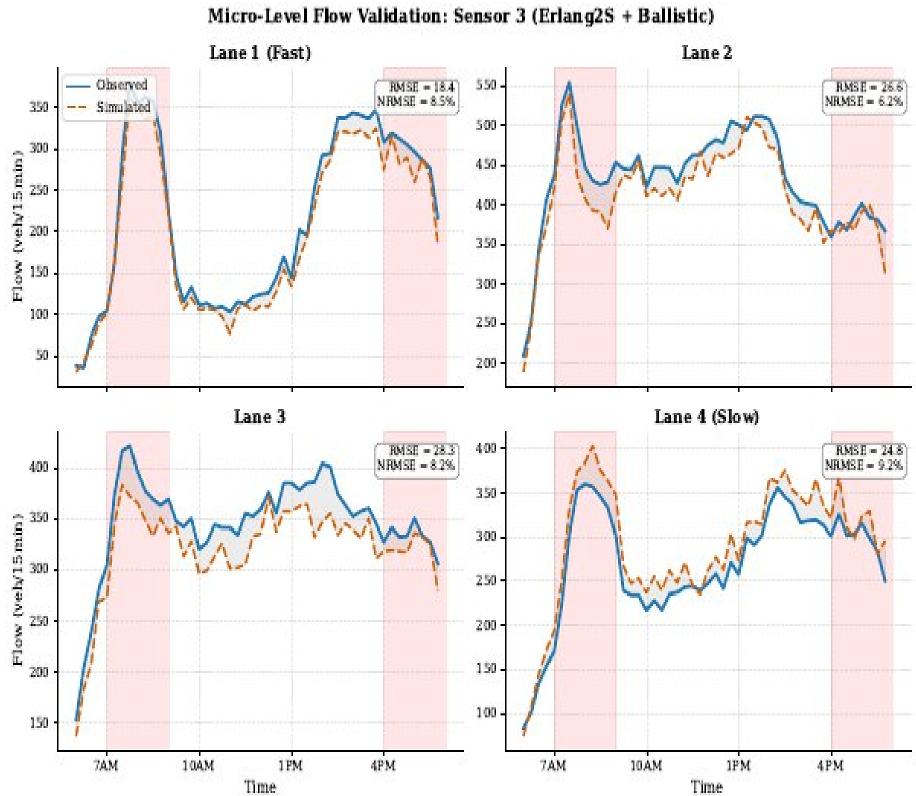
Fast lanes (L1):

- Highly predictable
- Flow NRMSE: 5.8% – 10.3%

Slow lanes (L4):

- High variability
- Affected by merging traffic
- Stop-and-go dynamics

Aggregate metrics hide these differences!



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Conclusions

Three key findings:

1. **Numerical integrator:** Negligible impact (<1%)
   → Use Ballistic (simplest, 3× faster)
2. **Arrival process:** Major impact (28% improvement)
   → Use Shifted Erlang-2 with minimum headway
3. **Lane-level validation:** Reveals hidden dynamics
   → Don't rely only on aggregate metrics

Future work: IDM calibration using 8 optimizers including SPSA with Momentum (SPSA-Mo), Analysis and result of each optimizers using individual lane level validation Metric. (Speed and flow per lane per interval)

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Questions?